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An Alternative Method for Edge Detection Using Schwarzian Derivatives

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Abstract

Edge detection techniques play a critical role in various image processing applications. In this study, a novel edge detection method is proposed by integrating traditional Sobel, Prewitt, and Canny operators with Schwarzian derivatives. Based on a review of the relevant literature, the proposed method appears to be the first edge detection filter that incorporates Schwarzian derivatives. Experimental results demonstrate that the proposed algorithm delivers performance comparable to widely used filters such as Sobel, Prewitt, and Canny, while offering a new perspective on edge detection in the field of image processing.

1. Introduction

In digital images, edges represent the boundaries that distinguish objects or regions, characterized by significant variations in brightness or color intensity. These transitions, where intensity levels change abruptly, are essential for defining the geometric and structural properties of objects. Edge detection is a fundamental image processing technique that aims to automatically identify these boundaries by detecting intensity discontinuities. This method detects abrupt variations in light intensity among neighboring pixels, making it essential for revealing object characteristics and contributing to the structural analysis of objects in an image. As a key component in object recognition and segmentation, edge detection enables accurate feature extraction, contributing to applications such as autonomous navigation, medical diagnostics, and remote sensing [1].

Over the years, numerous edge detection methods have been developed, each presenting unique benefits and constraints. Among these, certain approaches have stood out as classical methods and are now extensively used in the field. One such method, Sobel gradient operator, on the other hand, identifies edges by applying horizontal and vertical gradient filters of size 3×3 or 5×5 . By analyzing the intensity variations across the image pixels, this method allows for the estimation of both edge direction and magnitude [2]. The Prewitt gradient operator, utilizes static 3×3 filters to highlight horizontal and vertical edges. It computes gradients in both directions, determines their magnitudes, and extracts edge points accordingly [3]. Another conventional technique, the Canny edge detector, follows a multi-step process involving noise reduction, gradient calculation, non-maximum suppression, and edge tracking through double-thresholding. This enables the extraction of fine, well-defined edges while minimizing noise interference [4]. Over time, these traditional edge detection methods have undergone various refinements, leading to multiple improved versions. In this context, several studies have introduced enhancements such as the 'Improved Sobel Method' [5], 'Improved Prewitt Method' [6], and 'Improved Canny Method' [7]. Additionally, deep learning-based techniques have emerged as powerful alternatives, demonstrating significant advancements in edge detection across applications like autonomous navigation and medical image processing [1], [8].

The Schwarzian gradient operator, introduced by Hermann Schwarz, offers a perception-based approach to analysis by considering both first- and second-order derivative information together, without the need for any masking method. This improves the accuracy of edge detection

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while maintaining strong performance in preserving high-frequency details. Schwarzian derivatives have made significant contributions to the literature by being explored across various disciplines from past to present. For instance, they were used by Pizarro et al. to more accurately model the geometric distortion between images, providing a new perspective on warp methods [9]. In addition, they have been effectively applied in various fields such as signal analysis [10], system stability and dynamical systems [11], nonlinear differential equations [12]. Motivated by these insights, this paper introduces a novel method as an alternative to existing edge detection techniques. The proposed approach aims to provide a new perspective on edge detection by incorporating Schwarzian derivatives, which are integrated with traditional edge detection filters. In this framework, Otsu's thresholding algorithm, which is frequently employed for binarizing images, is used as a preprocessing step [13]. With its parameter-free and computationally efficient structure, it enables successful edge detection without requiring prior information.

This paper is organized as follows: Section II presents the mathematical background and includes the definitions of Schwarzian derivatives and the Otsu thresholding method. Section III describes in detail the dataset, the performance evaluation metrics and the proposed method. Finally, Section IV discusses the results and the general conclusions.

2. Mathematical Background

The proposed edge detection algorithm is based on the Schwarzian derivative. The Schwarzian derivative allows to study the geometric properties of a function and is an important mathematical tool for analyzing the behavior of analytic functions, especially in complex analysis. This derivative plays a critical role in understanding the conformal (angle-preserving) structure of the function and is especially used to investigate transformation properties in the complex plane. A proper definition of the Pre-Schwarzian derivative is required before moving on to the computation of the second-order Schwarzian derivative. To this end, let f be a locally bijective analytic function defined on $\Omega \subset \mathbb{C}$. The Pre-Schwarzian derivative serves as a preliminary step in the computation of the Schwarzian derivative, offering insights into the function's differential structure. The Pre-Schwarzian derivative of the function f is expressed as [14].

$$Pf = \frac{f''}{f'} \quad (2.1)$$

Furthermore, the second order Schwarzian derivative of the function f is defined as follows [15].

$$\begin{aligned} Sf &= (Pf)' - \frac{1}{2}(Pf)^2 \\ &= \left(\frac{f''}{f'}\right)' - \frac{1}{2}\left(\frac{f''}{f'}\right)^2 \end{aligned} \quad (2.2)$$

It is known that the Pre-Schwarzian derivative remains unchanged under Möbius transformations and vanishes if and only if f is linear. In contrast, the Schwarzian derivative equals zero solely for Möbius transformations [14]. This property allows the Schwarzian derivative to be used as a measure of how close functions are to Möbius transformations. Moreover, Schwarzian derivatives are widely used in many different mathematical fields, especially in conformal mapping, differential geometry and dynamical systems.

Various preprocessing techniques have been applied to improve the accuracy of edge detection. In this context, the widely used Otsu method is preferred for thresholding grayscale images. The method divides the image into two classes, background and foreground, and determines the threshold value that minimizes the total variance [13].

3. Proposed Method

The method developed here is important as it gives a new perspective by using Schwarzian derivatives for the first time in edge detection algorithm. A black and white or color image is taken as input to the algorithm, and if the image is in color, it is converted into a grayscale image. Then, the optimal threshold value based on the histogram is determined with the Otsu thresholding method. Using this threshold, the image is converted to binary format, where black pixels are represented as 0 and white pixels as 1. In this study, an operator $\mathcal{L}$ inspired by the Pre-Schwarzian derivative is defined as follows

$$\mathcal{L}I = \frac{I''}{\|I\|_F + I'} \quad (3.1)$$

Here, I represents the binarized form of the input image, and the Frobenius norm $\|I\|_F$ is introduced in the denominator to prevent potential division by zero issues that may arise in the computation of Schwarzian derivatives. Similarly, the operator $\mathcal{M}$ can be defined utilizing the second-order Schwarzian derivative as follows

$$\begin{aligned} \mathcal{M}I &= (\mathcal{L}I)' - \frac{1}{2}(\mathcal{L}I)^2 \\ &= \left(\frac{I''}{\|I\|_F + I'}\right)' - \frac{1}{2}\left(\frac{I''}{\|I\|_F + I'}\right)^2 \end{aligned} \quad (3.2)$$

In this study we develop a new algorithm to detect edge using Pre-Schwarzian and second-order Schwarzian derivatives. For this purpose we define two new operators that we apply to the images. When applying the operators, instead of the derivative terms in the operators, three different gradient-based edge detection methods are used such as Sobel, Prewitt, and Canny. For example, the Canny filter is applied once for the first-order term and twice for the second-order term before the results are substituted into the formula. As a result, three new algorithms

are developed, namely Schwarzian-Sobel, Schwarzian-Prewitt, and Schwarzian-Canny. These methods preserve the geometric sensitivity of the Schwarzian framework while integrating the directional strength and noise robustness of classical filters. The proposed approaches not only stabilize the computation, especially around regions with weak gradients, but also improve visual clarity and sharpness in the extracted edges. This integration constitutes to the core contribution of the study, offering an enriched mathematical structure for edge detection tasks.

4. Experimental Results and Discussion

To evaluate the performance of the algorithm, three widely used metrics in image processing —Peak Signal-to-Noise Ratio (PSNR), Mean Squared Error (MSE), and Structural Similarity Index (SSIM)— are employed. PSNR quantifies image distortion by comparing the peak signal power to the background noise level [16]. MSE calculates the average squared error between corresponding pixels of the reference and processed images, offering a straightforward measure of deviation [17]. SSIM evaluates perceptual similarity by incorporating structural, contrast, and luminance components [16]. These metrics collectively enable both numerical and perceptual evaluation of image quality and filter efficiency. The algorithm is implemented in MATLAB R2023b on a MacBook Air with an Apple M2 chip.

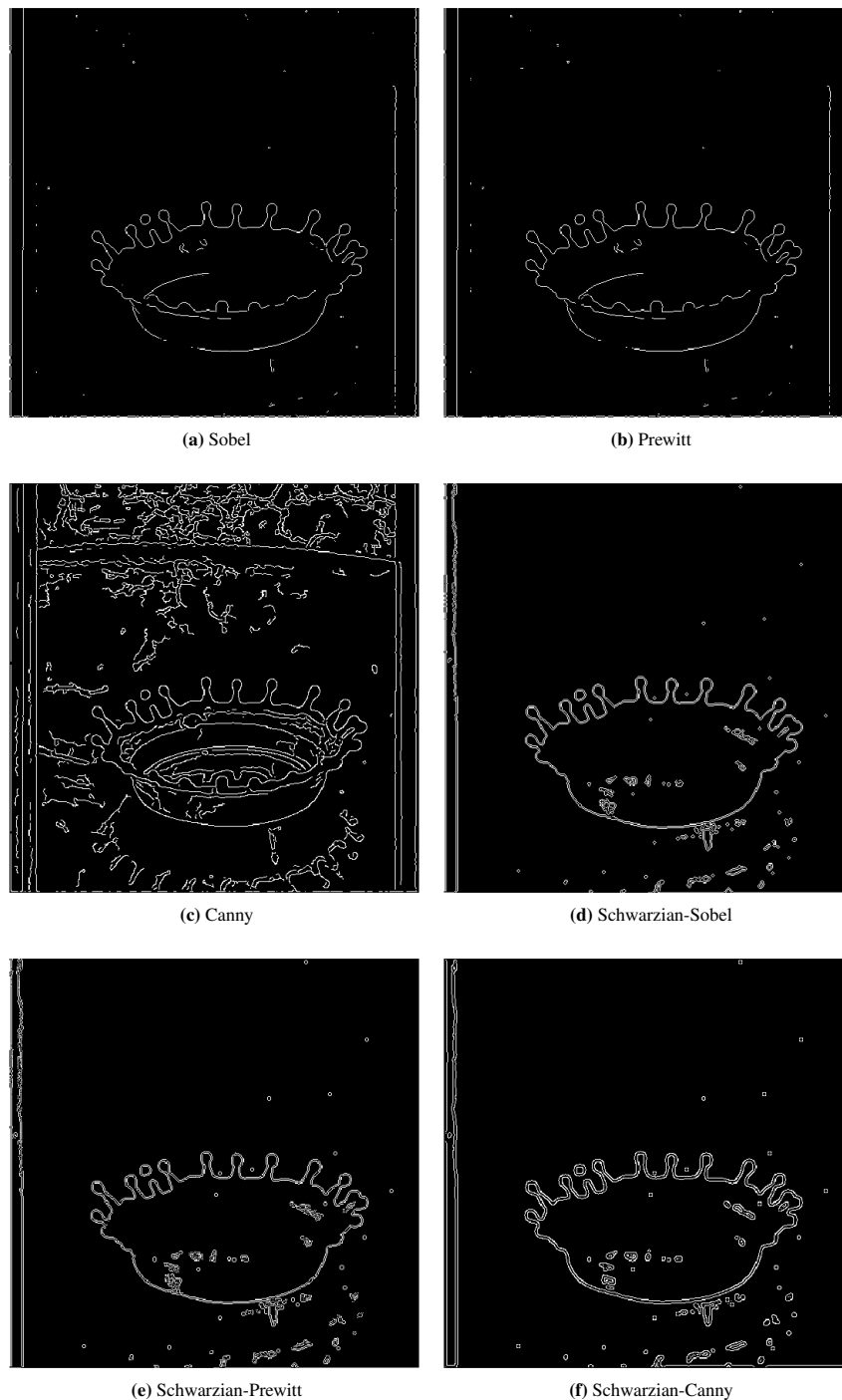


Figure 4.1: Splash image obtained using different filters.

For a comprehensive analysis of the method's performance under different conditions, test images of various resolutions from different sources are used. The Splash image with a resolution of 512×512 pixels from the Berkeley dataset [18] is selected. In addition, the Lena and Peppers images, each with a resolution of 512×512 pixels, are chosen from the standard_test_images dataset [19]. Among the test images taken from Gonzalez's textbook [20], there are Rear of a Vehicle (566×517 pixels), Building (1114×834 pixels), and Coins (300×246 pixels).

Figure 4.1 presents the edge detection results obtained from three classical methods (Sobel, Prewitt, Canny) and their corresponding Schwarzian-based variants (Schwarzian-Sobel, Schwarzian-Prewitt, Schwarzian-Canny) on the Splash image. While the Sobel and Prewitt methods are able to detect the outer contours of the Splash, they produce weak and discontinuous edges in detailed areas. The Canny method reveals more edges, but also introduces background noise and double-edge artifacts. In contrast, the proposed Schwarzian-based methods yield more defined and continuous edges, particularly enhancing clarity and contrast along the contour of the splash. Among them, the Schwarzian-Canny method preserves more fine details with reduced noise compared to its classical version. Overall, the Schwarzian-enhanced techniques generate sharper and more stable edge maps, especially in complex or low-contrast regions, outperforming traditional approaches in visual quality.

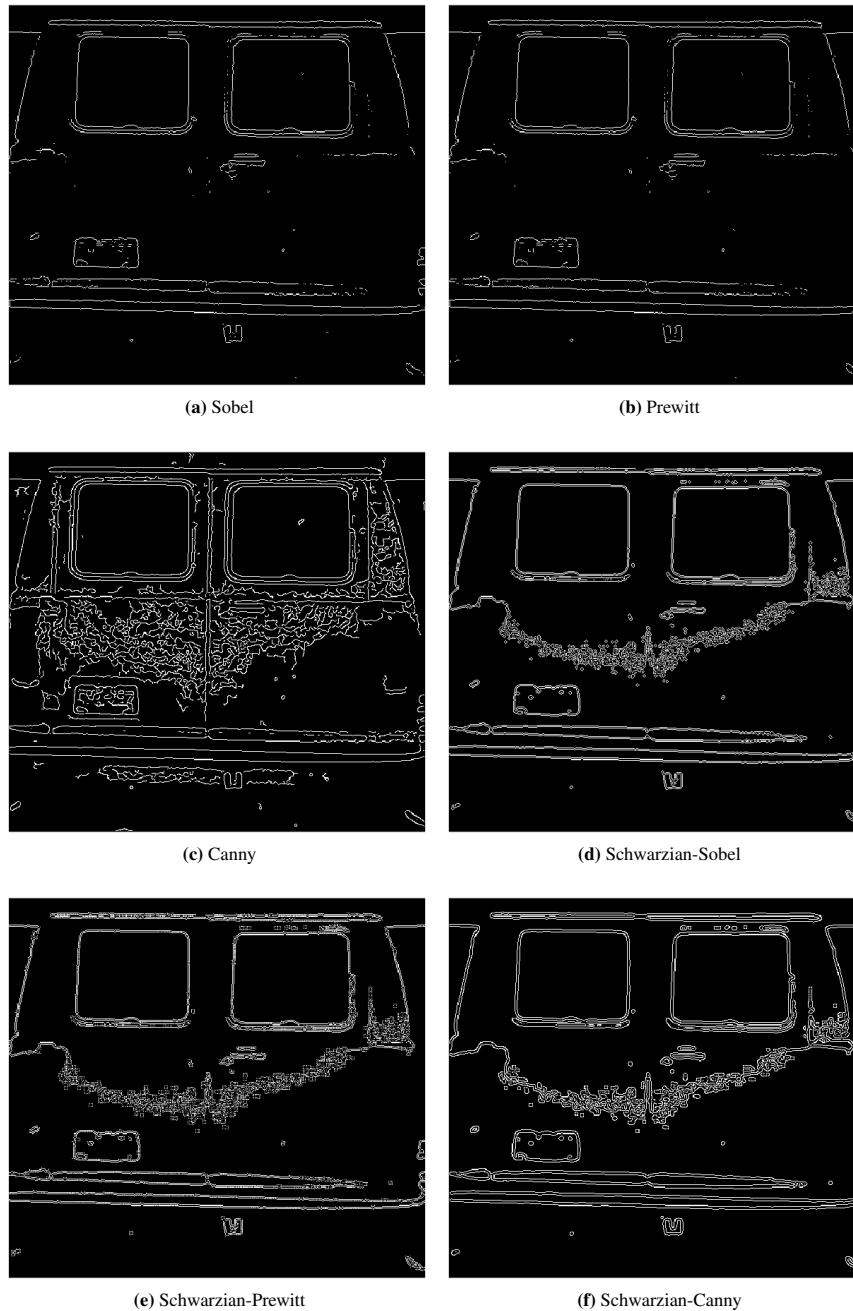


Figure 4.2: Rear Of a Vehicle image obtained using different filters.

Using the Rear of a Vehicle, Building, and Coins images, classical edge-detection algorithms were compared with their Schwarzian-enhanced counterparts, and the visual results are presented in Figures 4.2 - 4.4. In Figure 4.2, the Schwarzian-Canny method produces cleaner, more continuous edges, reduces noise, and preserves structural details better than the other approaches. For the Building image in Figure 4.3, Schwarzian-Prewitt and Schwarzian-Canny produce noticeably sharper roof lines and window boundaries, whereas Sobel and Prewitt give

weaker responses and the classical Canny introduces considerable noise; this confirms that, in low-contrast regions, Schwarzian-based methods deliver sharper and more balanced edge maps. Finally, in Figure 4.4, the classical Canny extracts many interior details of the coins but generates noisy edge patterns, while Schwarzian-Canny and Schwarzian-Sobel provide clean inner contours with minimal noise. Together, these observations demonstrate that the proposed Schwarzian-based algorithms consistently generate sharper and more stable edge maps than traditional techniques, especially in challenging or low-contrast scenes.

The quantitative results in Table 1 strongly support the visual improvements. In terms of SSIM, the Schwarzian-Sobel method achieves the highest score for every test image, noticeably outperforming the classical filters in structural similarity. For PSNR and MSE, the conventional Canny (Rear of a Vehicle, Coins) or Prewitt (Peppers, Building) lead in some instances; however, for the Splash image, the best performance across SSIM and PSNR/MSE is delivered by the Schwarzian-based approaches—particularly Schwarzian-Prewitt and Schwarzian-Sobel. In summary, the algorithms enriched with Schwarzian derivatives provide a substantial improvement in structural similarity while remaining competitive in PSNR and MSE, indicating that the enhancement in edge clarity is even greater than numerical metrics alone might suggest.

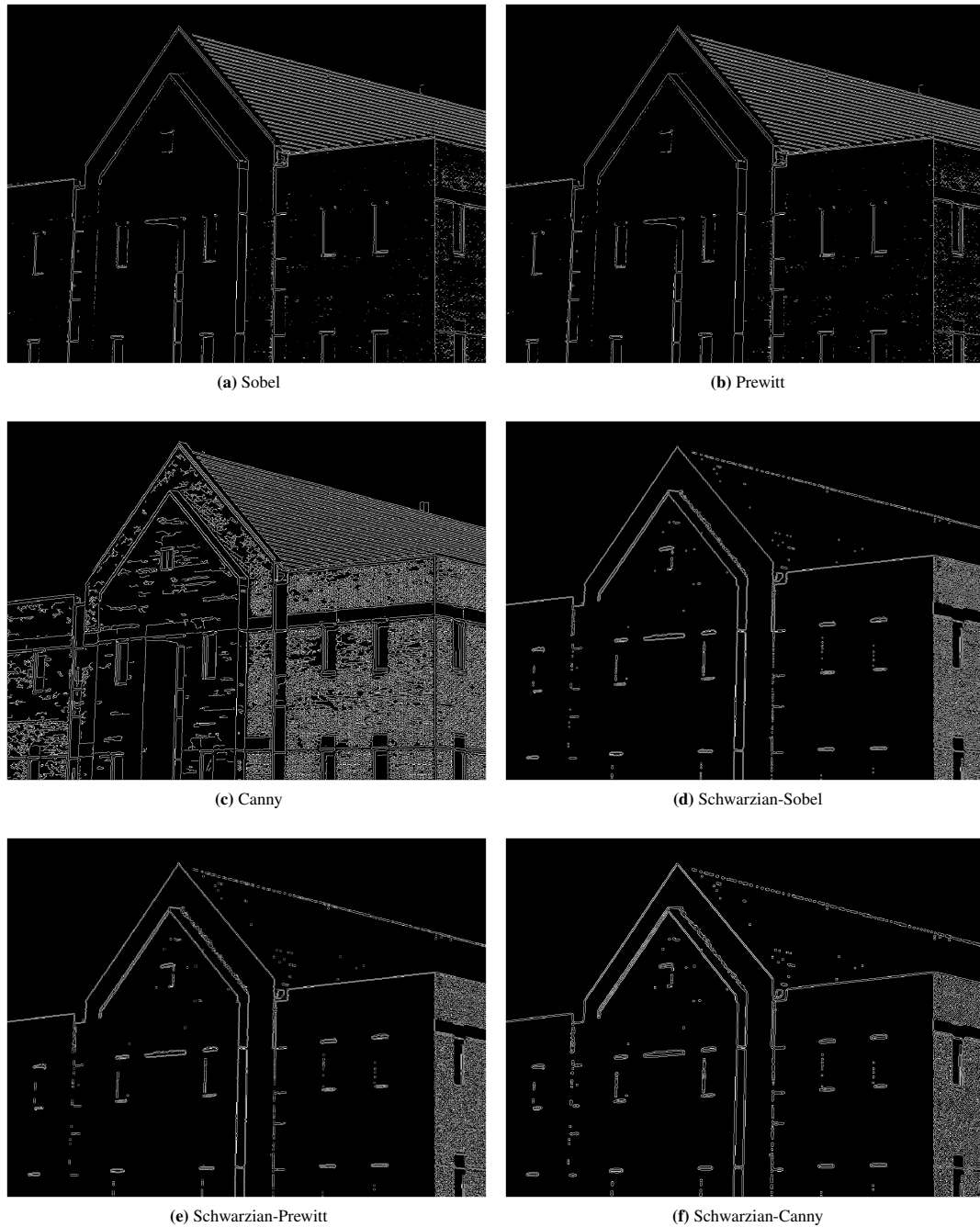


Figure 4.3: Building image obtained using different filters.

Although the proposed Schwarzian-Canny method produces more distinct and continuous edges in terms of visual quality, the PSNR and MSE values are lower than those of the classical Canny method for some images. This is particularly evident in the comparisons made on the Splash and Rear of a Vehicle images. The main reason for this is that both PSNR and MSE metrics are sensitive only to pixel-level differences and fail to adequately reflect structural improvements related to visual perception. In other words, although edge clarity, continuity, and structural integrity have been improved, these enhancements are not always directly reflected in numerical metrics. In this context, the SSIM

metric has proven to be a more consistent indicator and has been observed to better represent perceptual image quality.

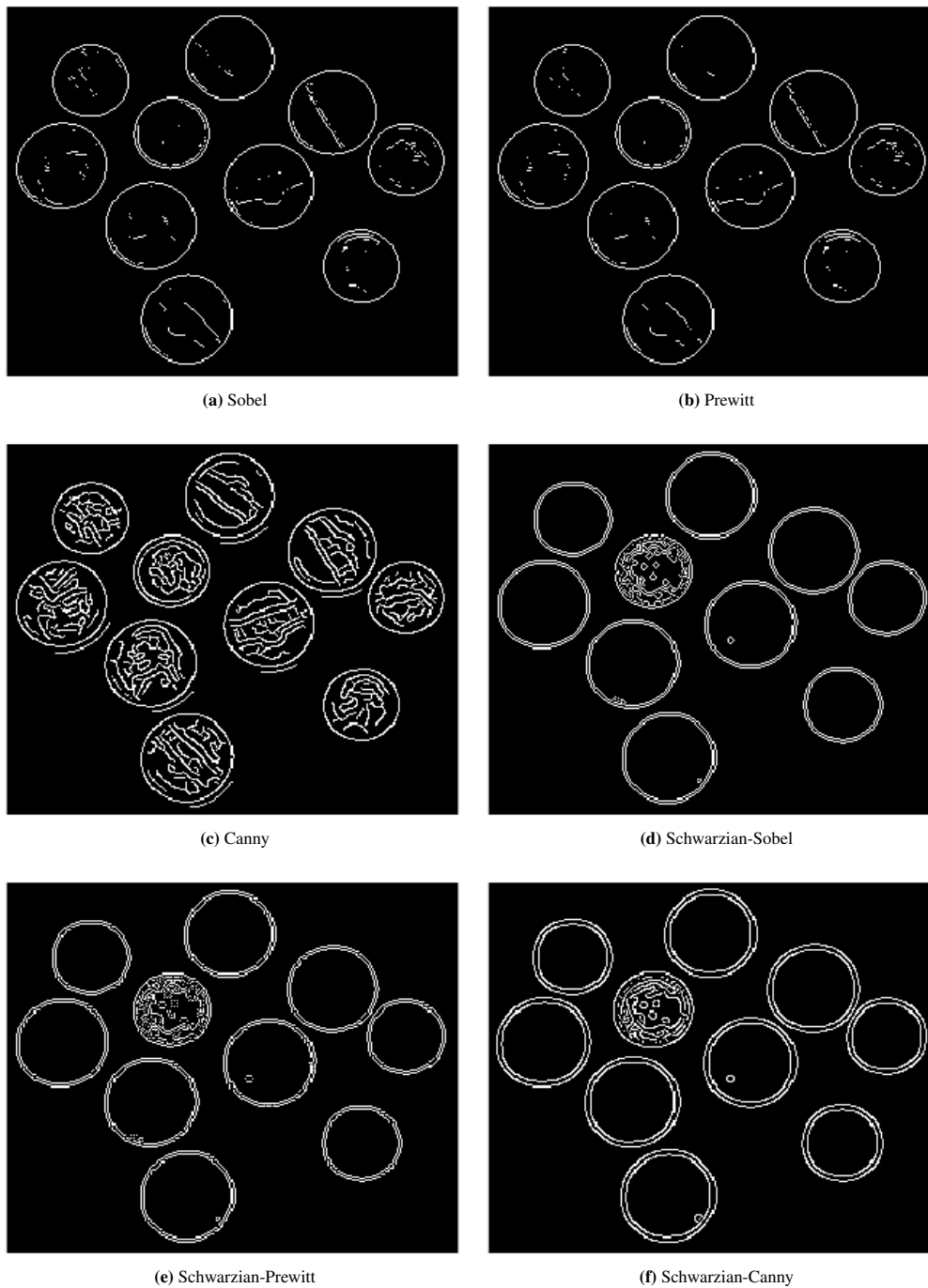


Figure 4.4: Coins image obtained using different filters.

5. Conclusion

In this study, a new edge detection algorithm has been developed using Schwarzian derivatives. Two new operators have been defined for the development of the algorithm and three different edge detection methods such as Sobel, Prewitt and Canny have been used instead of the derivative terms themselves while applying these operators. The algorithms produced in this way are named Schwarzian-Sobel, Schwarzian-Prewitt and Schwarzian-Canny, respectively. Numerical and visual performances of these methods are presented as tables and figures calculated using different images. Experimental results (Table 1, Figures 4.1 - 4.4) indicate that the proposed methods perform competitively or even surpass classical filters in terms of SSIM scores. Overall, the Schwarzian-enhanced algorithms generate sharper, more continuous edge maps, especially in low-contrast or visually complex scenes, confirming their advantage over traditional techniques.

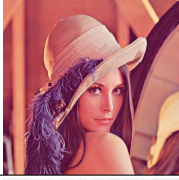
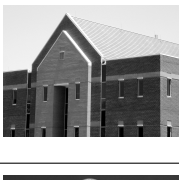
Image	Method	SSIM	PSNR	MSE
	Sobel	0.009430	6.898337	0.204252
	Prewitt	0.009414	6.897955	0.204270
	Canny	0.009797	6.615345	0.218005
	Schwarzian-Sobel	0.742205	7.211641	0.190036
	Schwarzian-Prewitt	0.739043	7.220750	0.189638
	Schwarzian-Canny	0.731208	7.161151	0.192258
	Sobel	0.006871	5.578296	0.276803
	Prewitt	0.006447	5.577212	0.276872
	Canny	0.003802	5.467223	0.283973
	Schwarzian-Sobel	0.121158	1.519941	0.704703
	Schwarzian-Prewitt	0.121137	1.527917	0.703410
	Schwarzian-Canny	0.101987	1.470594	0.712755
	Sobel	0.007094	6.155226	0.242369
	Prewitt	0.006969	6.155399	0.242360
	Canny	0.009405	6.236686	0.237865
	Schwarzian-Sobel	0.509129	4.286817	0.372665
	Schwarzian-Prewitt	0.505910	4.339081	0.368207
	Schwarzian-Canny	0.493007	4.244699	0.376296
	Sobel	0.016735	4.900166	0.323581
	Prewitt	0.017241	4.907620	0.323026
	Canny	0.017062	4.851551	0.327224
	Schwarzian-Sobel	0.264698	2.576716	0.552495
	Schwarzian-Prewitt	0.262640	2.583506	0.551632
	Schwarzian-Canny	0.231504	2.474592	0.565641
	Sobel	0.007953	4.443984	0.359419
	Prewitt	0.008149	4.444206	0.359401
	Canny	0.013527	4.053251	0.393256
	Schwarzian-Sobel	0.504904	3.677952	0.428751
	Schwarzian-Prewitt	0.500346	3.664477	0.430083
	Schwarzian-Canny	0.495896	3.683867	0.428167
	Sobel	0.009153	6.806098	0.208636
	Prewitt	0.008407	6.794439	0.209197
	Canny	0.007873	7.205962	0.190285
	Schwarzian-Sobel	0.535291	4.829339	0.328902
	Schwarzian-Prewitt	0.532056	4.831010	0.328775
	Schwarzian-Canny	0.514411	4.766006	0.333733

Table 1: Performance Metrics for Different Methods

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References

- [1] C. Kim, M. Jang, Y. Han, Y. Hong, and W. Lee, *Skin lesion classification using hybrid convolutional neural network with edge, color, and texture information*, Appl. Sci., **13**(9) (2023), Article ID 5497, doi:10.3390/app13095497.
- [2] I. Sobel and G. Feldman, *A 3x3 isotropic gradient operator for image processing*, Stanford Artificial Intelligence Project, Stanford University, 1968.
- [3] J. M. S. Prewitt, *Object enhancement and extraction*, in *Picture Processing and Psychopictorics*, B. S. Lipkin and A. Rosenfeld (Eds.), Academic Press, New York, 1970, pp. 75–149.
- [4] J. Canny, *A computational approach to edge detection*, IEEE Trans. Pattern Anal. Mach. Intell., **8**(6) (1986), 679–698, doi:10.1109/TPAMI.1986.4767851.
- [5] J. Tang, N. Liu, and H. Jian, *An edge detection method based on improved Sobel operator for infrared target extraction*, in Proc. 5th IEEE Int. Conf. Power Intell. Comput. Syst. (ICPICS), Shenyang, China, 2023, pp. 119–123, doi:10.1109/ICPICS58376.2023.10235635.
- [6] S. Balochian and H. Balochian, *Edge detection on noisy images using Prewitt operator and fractional order differentiation*, Multimed. Tools Appl., **81** (2022), 9759–9770, doi:10.1007/s11042-022-12011-1.

- [7] C. Xu, J. Su, T. Wang, Y. Zhang, and A. Yu, *Improved Canny edge detection operator*, in Proc. 7th Int. Conf. Commun. Inf. Syst. (ICCIS), Chongqing, China, 2023, pp. 135–140, doi:10.1109/ICCIS59958.2023.10453881.
- [8] C.-i. Kim, J. Park, Y. Park, W. Jung, and Y.-s. Lim, *Deep learning-based real-time traffic sign recognition system for urban environments*, *Infrastructures*, **8**(2) (2023), Article ID 20, doi:10.3390/infrastructures8020020.
- [9] D. Pizarro, R. Khan, and A. Bartoli, *Schwarzs: Locally projective image warps based on 2D Schwarzian derivatives*, *Int. J. Comput. Vis.*, **119**(1) (2016), 93–109, doi:10.1007/s11263-016-0882-9.
- [10] R. M. Demirer and S. Kesebir, *The entropy of chaotic transitions of EEG phase growth in bipolar disorder with lithium carbonate*, *Sci. Rep.*, **11** (2021), Article ID 11888, doi:10.1038/s41598-021-91350-9.
- [11] D. Singer, *Stable orbits and bifurcation of maps of the interval*, *SIAM J. Appl. Math.*, **35**(2) (1978), 260–267, doi:10.1137/0135020.
- [12] D. Weiss, *The Painlevé property for partial differential equations. II: Bäcklund transformation, Lax pairs, and the Schwarzian derivative*, *J. Math. Phys.*, **24**(6) (1983), 1405–1413, doi:10.1063/1.525875.
- [13] N. Otsu, *A threshold selection method from gray-level histograms*, *IEEE Trans. Syst. Man Cybern.*, **9**(1) (1979), 62–66, doi:10.1109/TSMC.1979.4310076.
- [14] V. Bravo, R. Hernández, S. Ponnusamy, and O. Venegas, *Pre-Schwarzian and Schwarzian derivatives of logharmonic mappings*, *Monatshefte für Math.*, **199** (2022), 733–754, doi:10.1007/s00605-021-01659-w.
- [15] M. Chuaqui, J. Gröhn, and J. Rättyä, *Generalized Schwarzian derivatives and higher order differential equations*, *Math. Proc. Camb. Philos. Soc.*, **151**(2) (2011), 339–354, doi:10.1017/S0305004111000296.
- [16] A. Horé and D. Ziou, *Image quality metrics: PSNR vs. SSIM*, in Proc. 20th Int. Conf. on Pattern Recognition (ICPR), Istanbul, Turkey, 2010, pp. 2366–2369, doi:10.1109/ICPR.2010.579.
- [17] H. Marmolin, *Subjective MSE measures*, *IEEE Trans. Syst. Man Cybern.*, **16**(3) (1986), 486–489, doi:10.1109/TSMC.1986.4308985.
- [18] *Berkeley Segmentation Dataset and Benchmark*, EECS, University of California, Berkeley, [Online]. Available: <https://www2.eecs.berkeley.edu/Research/Projects/CS/vision/bsds/>
- [19] *Standard Test Images*, Image Processing Place, [Online]. Available: https://www.imageprocessingplace.com/root_files_V3/image_databases.htm
- [20] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed., Pearson, New York, 2018, ISBN: 978-0-13-335672-4.